Argo CD and Redis for K8s

American Airlines

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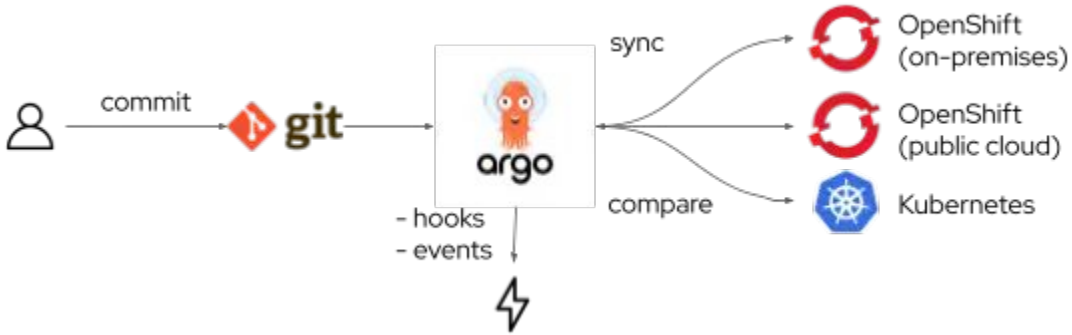
Argo CD

- Continuous Deployment tool
- A form of “Kubernetes of Kuberneteses”
- Git repo state → Kubernetes cluster
  - Basic unit: **Application**
  - You can have multiple applications in one namespace
- Optional auto-sync
- Web UI shows current state



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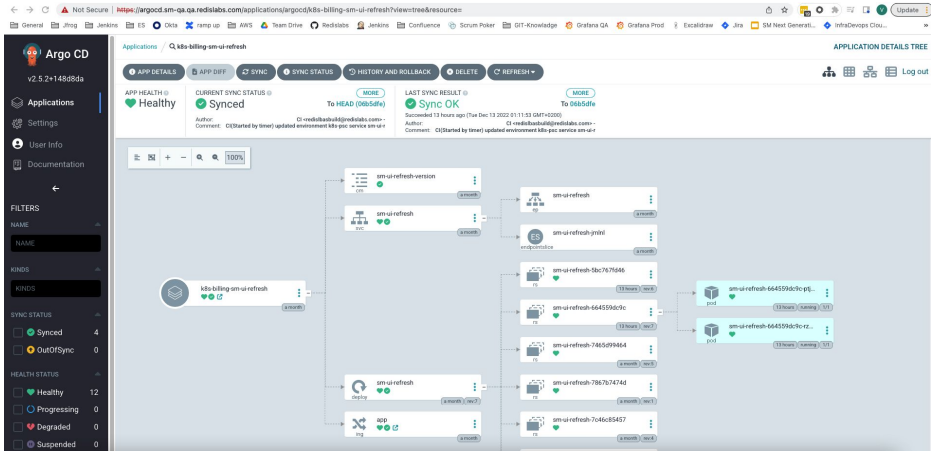
ArgoCD

ArgoCD is a Kubernetes-native continuous deployment tool, ArgoCD can pull updated code from Git repositories and deploy it directly to k8s resources, continuously monitoring all running applications and comparing their live state to the desired state specified in Git repository

ArgoCD UI



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ArgoCD Application

The application CRD is the Kubernetes resource object representing a deployed application instance in the environment.

```yaml
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: argo-demo
  namespace: argocd-namespace
spec:
  project: default
  source:
    repoURL: https://github.com/repo/name.git
    targetRevision: HEAD
    path: path/to/repo
  destination:
    server: https://kubernetes.default.svc
    namespace: rec-namespace
  syncPolicy:
    automated:
      prune: true
      selfHeal: true
```

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Application source/destination sections

Application CRD is defined by two key pieces of information

source - reference to desire state in Git (repository, revision, path, values.yaml)

```
source:
   repoURL: https://github.com/redislabsdev/cloud-apps-deployment.git
   targetRevision: HEAD
   path: environments/sm-qa/k8s-aa/sm-ui-refresh
   helm:
     releaseName: sm-ui-refresh
     passCredentials: true
     valueFiles:
       - values.yaml
```

destination - reference to target cluster and namespace where to deploy desired k8s manifests

```
destination:
   server: https://kubernetes.default.svc
   namespace: k8s-a
```

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ArgoCD Sync policy

ArgoCD has the ability to automatically sync an application when its detects difference between the desire manifests in Git, and life state in the cluster. A benefit of automatic sync is that CI/CD pipelines no longer need direct access to ArgoCD to perform the deployment

Options:

- **automated** - ArgoCD automatically sync the k8s state to be align with git changes
- **selfHeal** - ArgoCD will prevent manual changes on the cluster (like direct kubectl commands)
- **prune** - ArgoCD will automatically delete k8s object(s) that was dropped from git (by default is false)

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Helm

- A Kubernetes package manager
  - Similar to `pip`, `npm`, etc.
- `helm install` installs Kubernetes-based applications on a Kubernetes cluster
- "**Helm Charts**" define Kubernetes resources
  - They use the Go templating language
  - Templates are populated from **value files**
  - All value files merged in given order
  - Merger is recursive
    - You can override specific fields on their own
- Kustomize defines pure YAMLs
  - No templating code-based construction
  - The Kubernetes YAMLs are merged, rather than value files
  - Replacement is done through patches
    - Very limited and often difficult to work with

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Thank you.

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